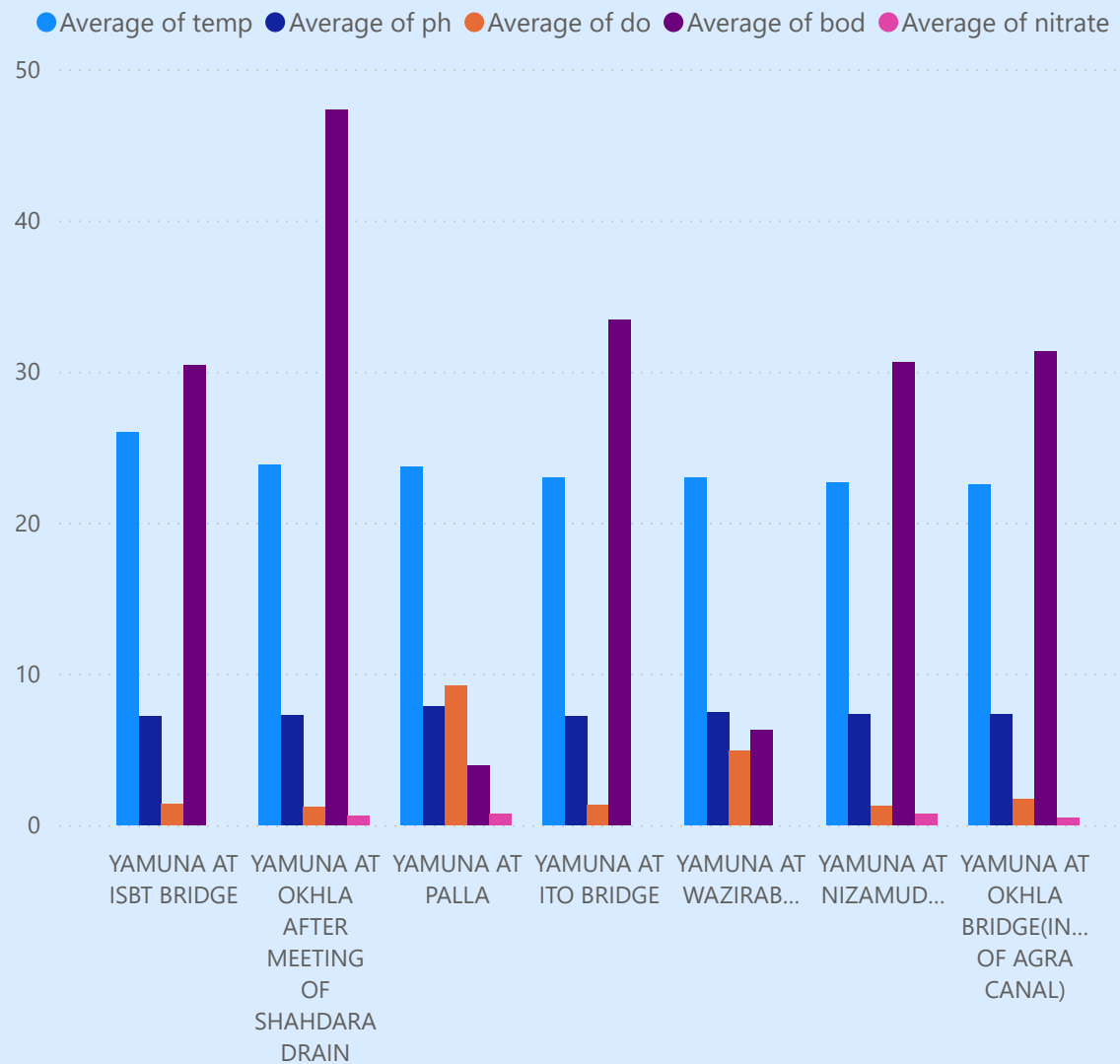


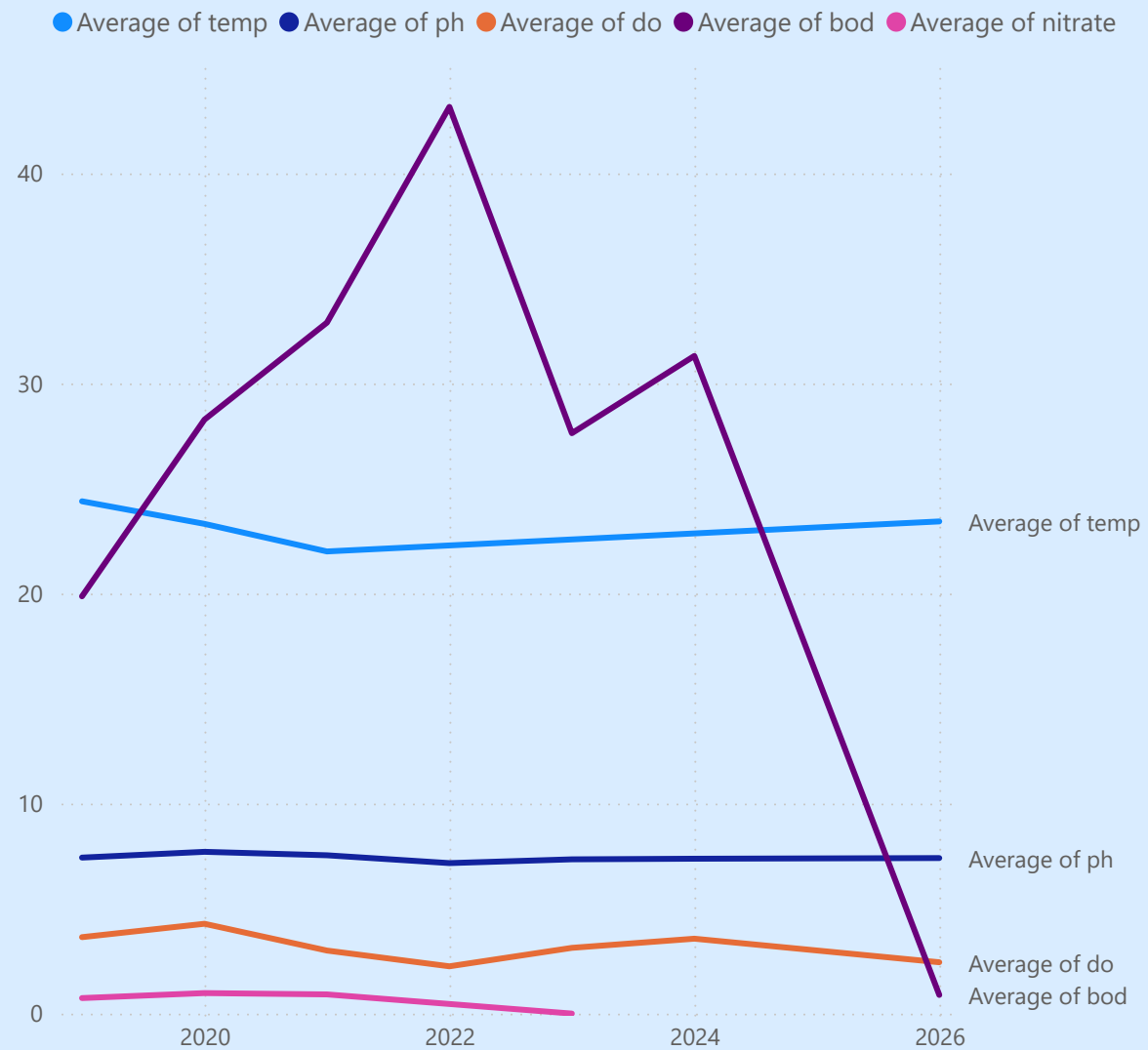
A person wearing a white lab coat and gloves is using a pipette to transfer liquid into a test tube. The background shows a river and some trees. The text is overlaid on a semi-transparent white box.

*Assessment of Yamuna River Water Quality using
CPCB Secondary data & Manually collected
samples by Power BI Dashboard and testing them
for further improvements*

Spatial variations



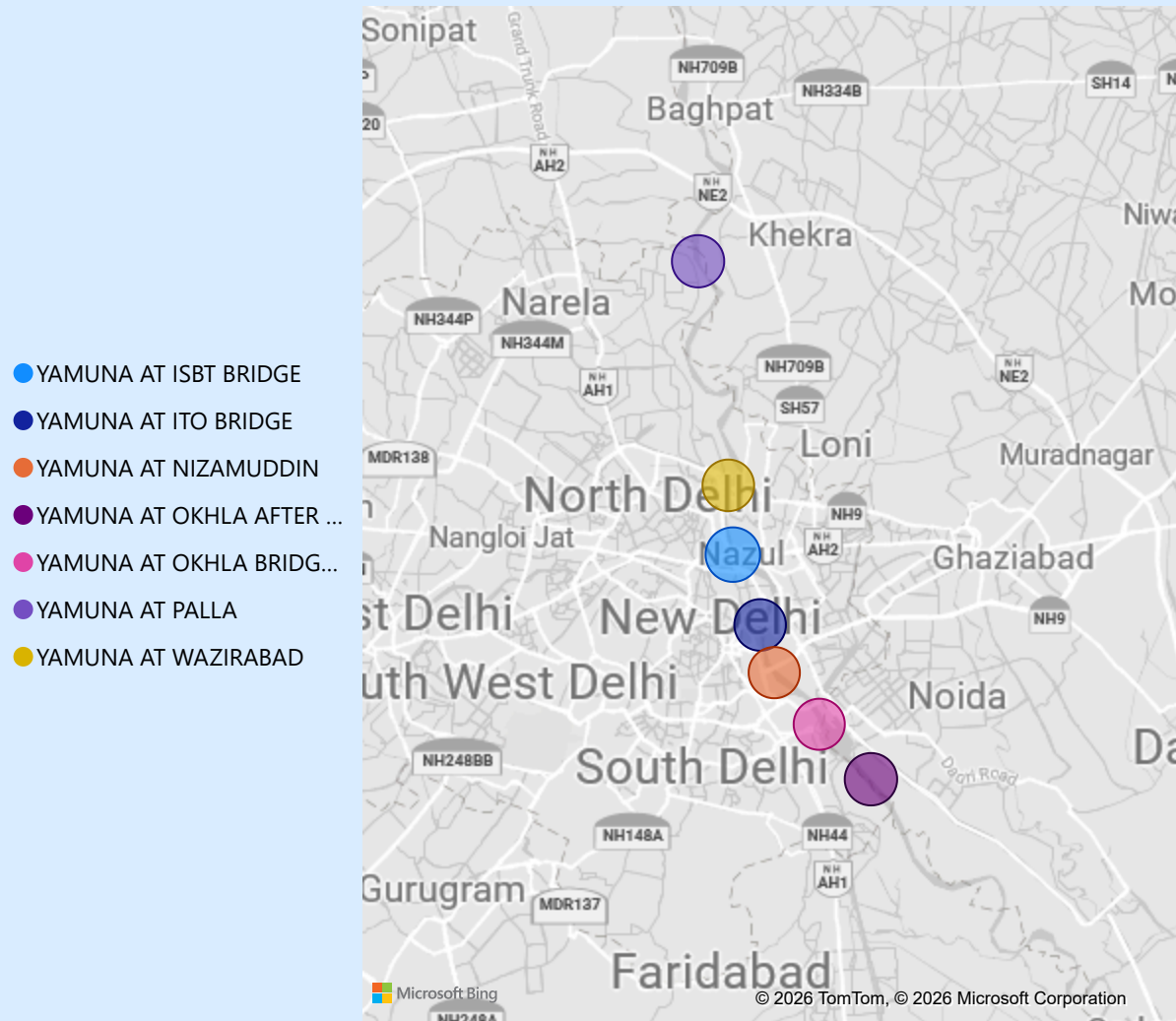
Temporal variations



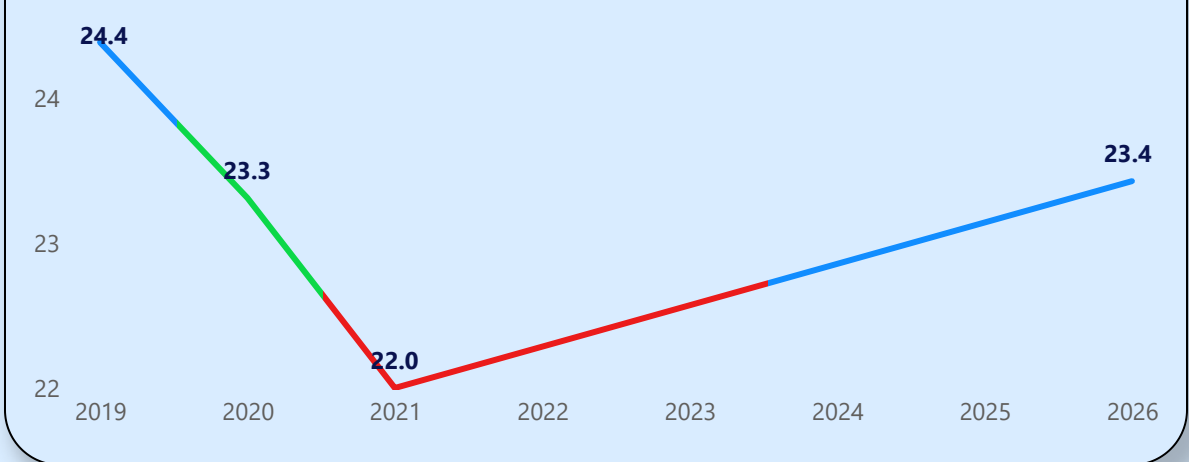


Temperature(°C)

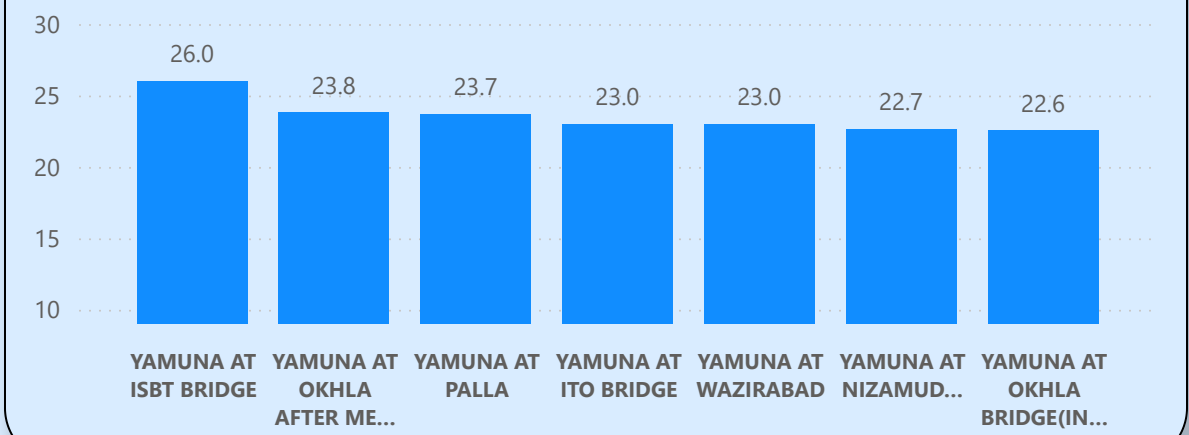
Temperature by location



Temperature by time



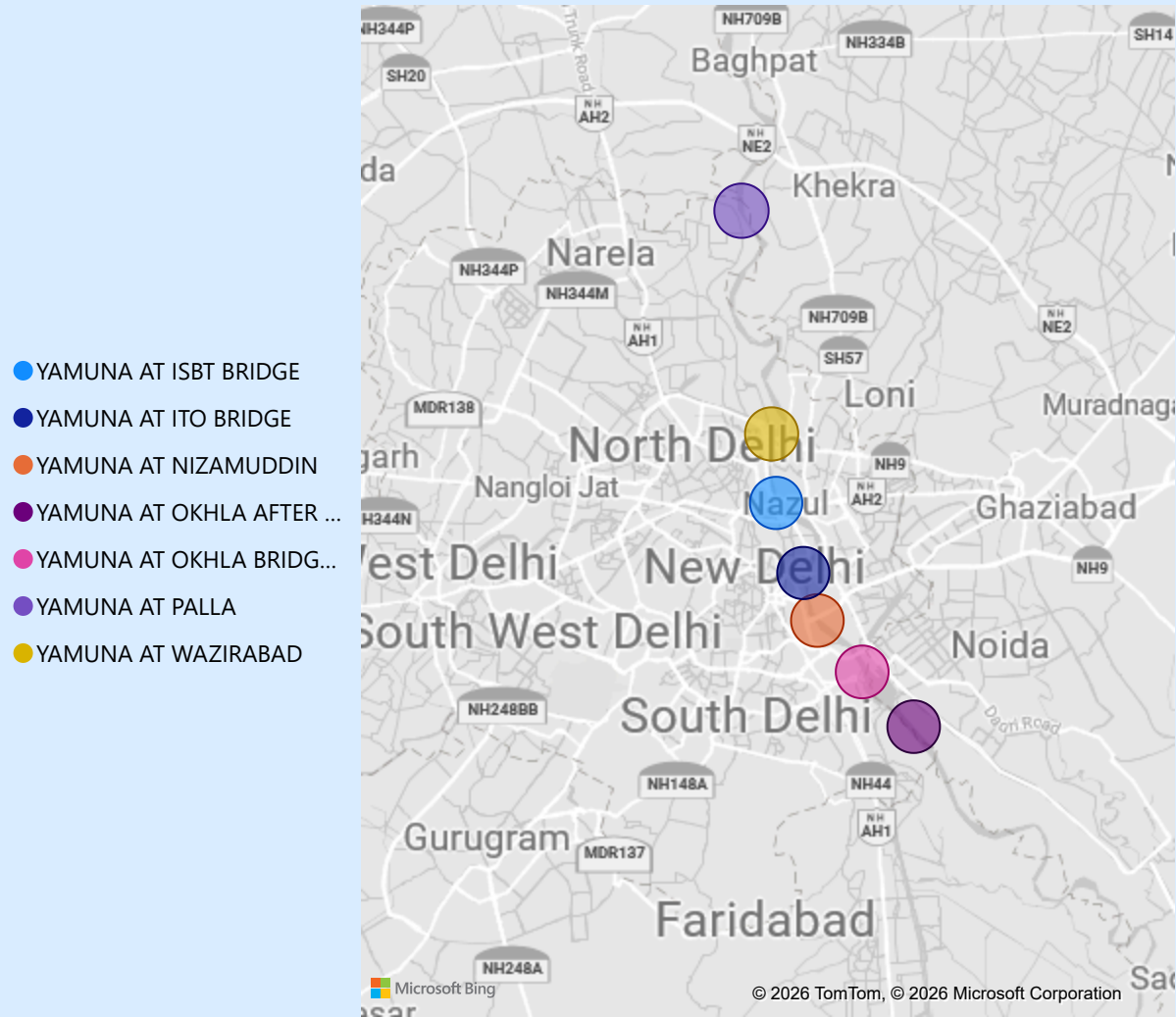
Temperature by location



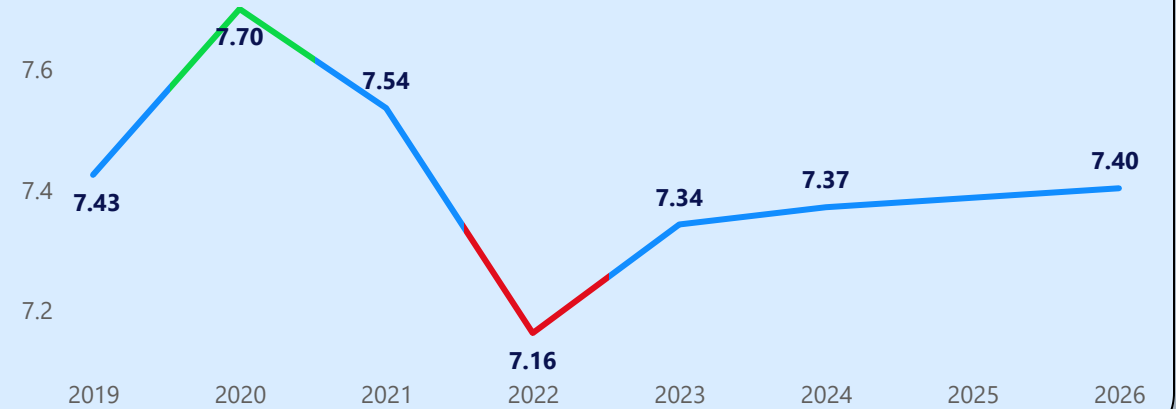


pH (6.5-8.5)

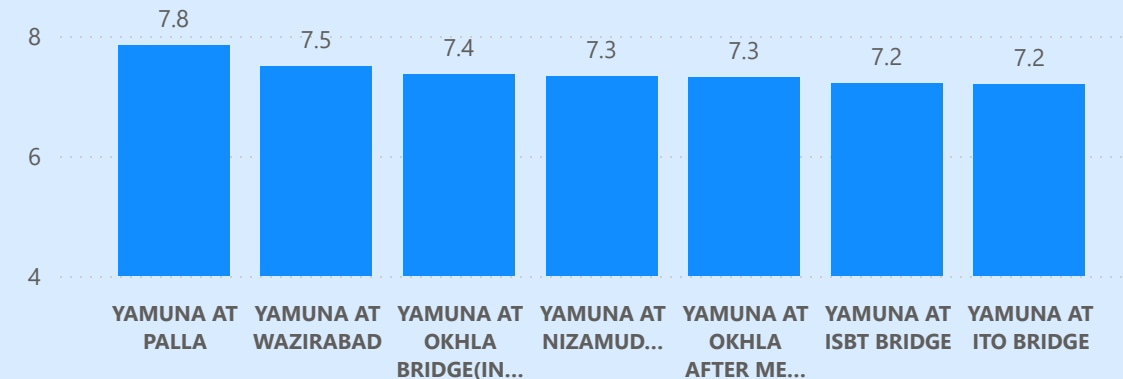
pH by location



pH by time



ph by location





The line chart illustrates the trend of the labour force in thousands from 2019 to 2026. The y-axis represents the number of people in thousands, ranging from 2 to 4. The x-axis shows the years from 2019 to 2026. The data points are as follows:

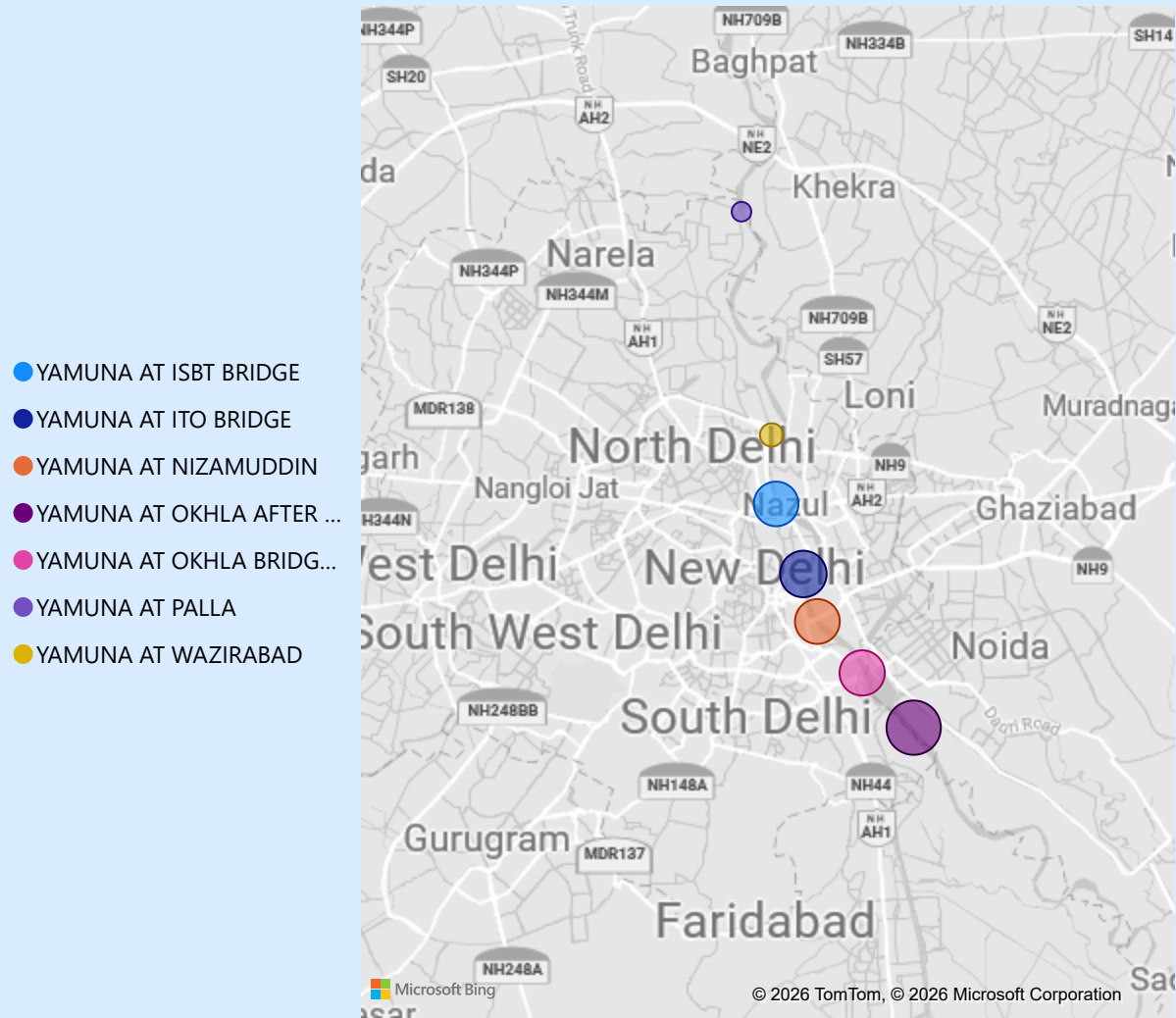
Year	Labour Force (thousands)
2019	3.6
2020	4.3
2021	3.0
2022	2.3
2023	3.1
2024	3.6
2025	3.0
2026	2.4

Location	Number of people
YAMUNA AT PALLA	9.2
YAMUNA AT WAZIRABAD BRIDGE(IN...)	5.0
YAMUNA AT OKHLA	1.8
YAMUNA AT ISBT BRIDGE	1.4
YAMUNA AT ITO BRIDGE	1.4
YAMUNA AT NIZAMUD... (Nizamuddin)	1.3
YAMUNA AT OKHLA AFTER ME...	1.2

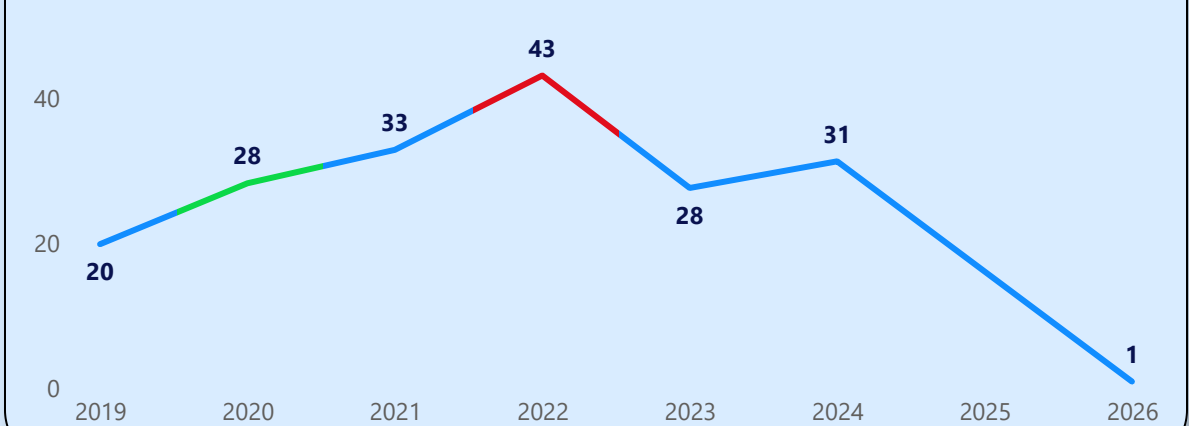


Biological Oxygen Demand (mg/L)

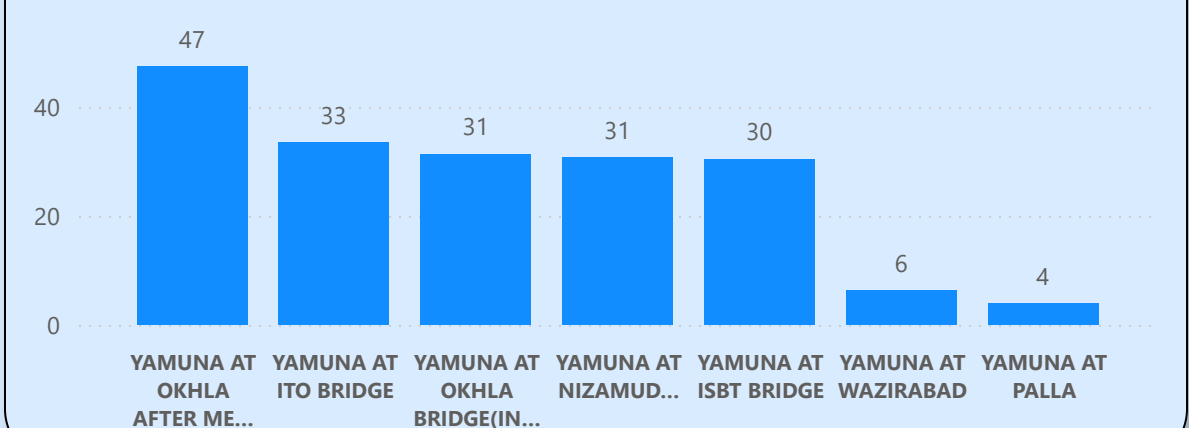
B.O.D. by location



B.O.D. by time



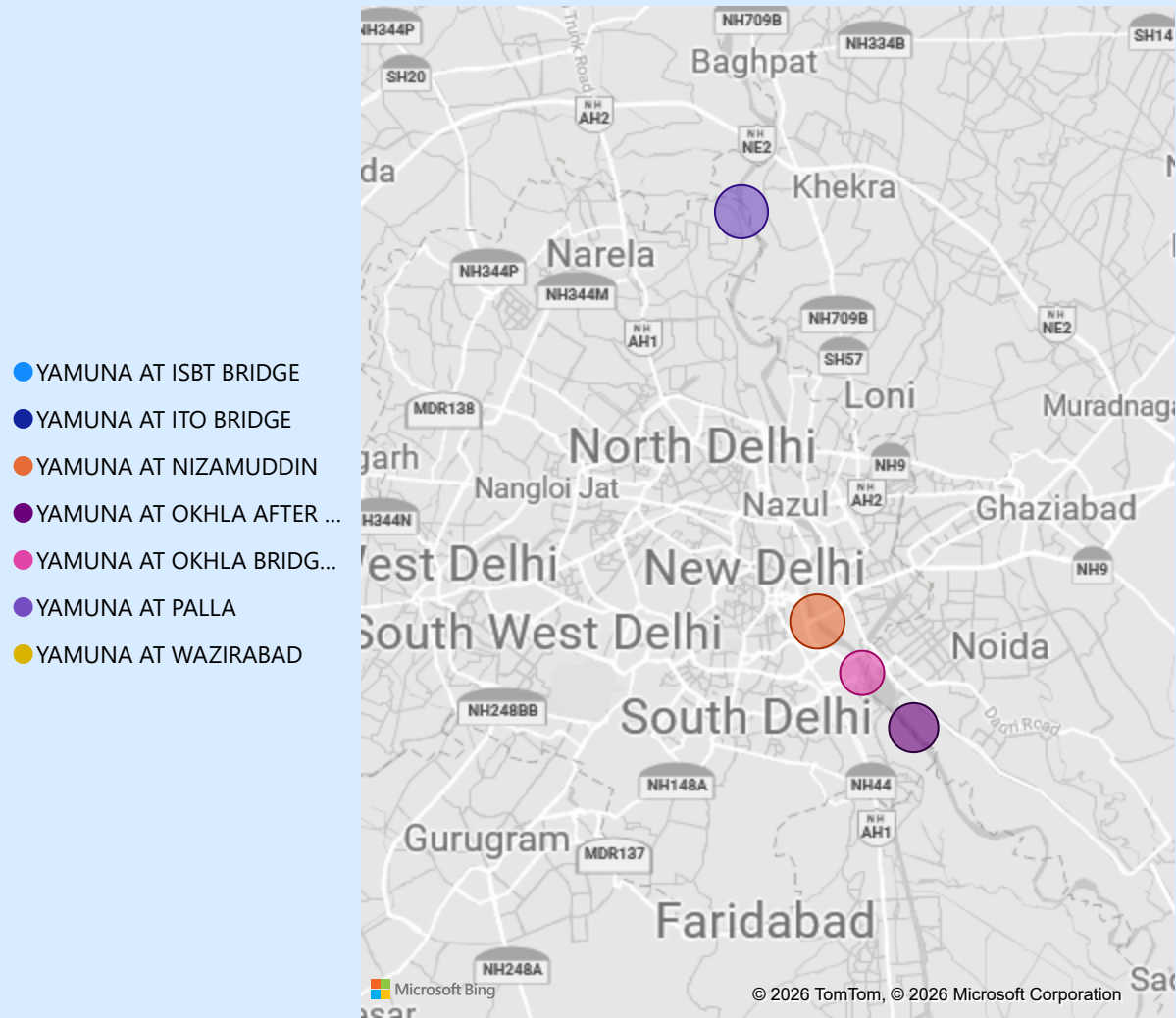
B.O.D. by location



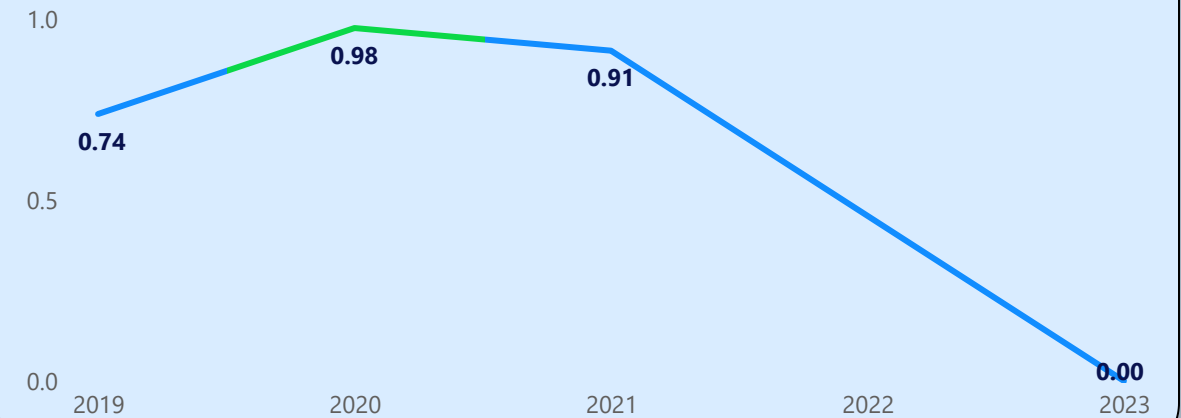


Nitrate (mg/L)

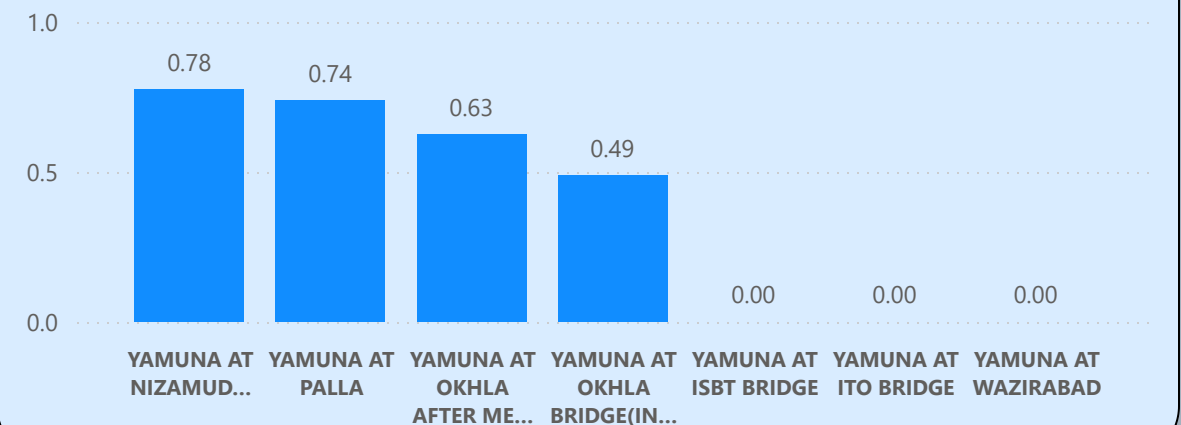
Nitrate. by location



Nitrate by time



Nitrate by location



Insights

Temporal Insights (Time-based)

Temperature

Temperature remains fairly stable between 22°C and 24°C over the years, indicating that it does not significantly impact water pollution levels.

pH

The pH values range between 7.1 and 7.7, which are within the acceptable range, suggesting that the water is chemically neutral and not affected by acidic or alkaline pollution.

Dissolved Oxygen (DO)

Dissolved Oxygen levels are consistently low, mostly below 5 mg/L, which indicates poor water quality and stress on aquatic life.

BOD

BOD levels are significantly high, reaching up to around 47 mg/L, which clearly indicates heavy organic pollution from sewage and waste discharge.

Nitrate

Nitrate levels are moderate and do not show critical pollution, suggesting that nutrient pollution is present but not the dominant issue.

Spatial Insights (Location-based)

Clean Zones

Locations like Palla and Wazirabad show better water quality, with higher dissolved oxygen and lower BOD, indicating relatively clean water at the entry point of the river.

Polluted Zones

As the river flows through Delhi, especially at ITO, Nizamuddin, and ISBT, water quality deteriorates sharply, with a significant drop in dissolved oxygen and rise in BOD.

Critical Zone

The most polluted location is Okhla after the meeting with Shahdara drain, where BOD is highest and dissolved oxygen is lowest, indicating severe pollution due to untreated wastewater discharge.

Relationship Insight

There is a strong inverse relationship between BOD and Dissolved Oxygen, meaning that as organic pollution increases, oxygen levels decrease, which is a clear indicator of deteriorating water quality.

Final Conclusion

In conclusion, the study highlights that the major cause of Yamuna pollution is untreated sewage and drain discharge in urban areas. Effective wastewater treatment and pollution control measures are essential to improve river water quality. The primary data collected in 2026 aligns with the historical CPCB trends, validating the reliability of the analysis and confirming that pollution remains a persistent issue.